

INFT-1205 – Data Communications & Networking II

Capstone Project

Floor 1 Switch Configuration

Institution: Durham College

By: Group 1

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Document Overview

This document serves as a brief documentation of the FL1 Switch. It includes an overview of the FL1 Switch, along with its configuration.

Device Overview

The FL1 switch is the main switch on Floor 1 of Datacore Solutions. It hosts VLANs for the IT, Customer Support, and Sales and Marketing departments. It connects to the Datacore Router via an 802.1Q trunk, and to FL2 via a 3-lane EtherChannel, configured using LACP. FL1 is fully configured with layer 2 security and includes a Parking Lot VLAN for unused ports.

Configuration

Building configuration...

Current configuration : 11256 bytes

!

! Last configuration change at 08:39:57 UTC Fri Mar 27 2026

!

version 17.6

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption

platform punt-keepalive disable-kernel-core

!

hostname FL1

!

!

vrf definition Mgmt-vrf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

enable secret 9

\$9\$eVvdtJchVtvYgE\$BVwwdulXLnX5jp6mDG9UGTp.AVujwRM9aJkDUt2la1w

!

no aaa new-model

switch 1 provision c9300l-24p-4g

!

!

!

!

!

!

!

!

!

no ip domain lookup

!

!

!

ip dhcp snooping vlan 10,20,30,40,50,60,70,80

ip dhcp snooping

ip arp inspection vlan 10,20,30,40,50,60,70,80

login on-success log

!

!

!

!

!
!
!
!

crypto pki trustpoint TP-self-signed-2873441000

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2873441000

revocation-check none

rsakeypair TP-self-signed-2873441000

!
!

crypto pki certificate chain TP-self-signed-2873441000

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32383733 34343130 3030301E 170D3236 30333236 31313030
35365A17 0D333630 33323531 31303035 365A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 38373334
34313030 30308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 0100EC48 8B13523E C2F357A6 F4ACC152 3890728D DDE0B44C 68CAB616
A1696C5F EAA515B3 9350916D 2CD4BD3E A65293C8 46E56D95 A9C7692E 5AFA38DA
A865C4E2 E8259C9E DA77037B BB3B0CAA 73FA2794 A9EF6B9E 62218A22 7A5DAB03
ADE28230 B0A7CF07 72ECE09B 6922B5DD 532E1359 12173FB2 93BFBD82 11E5CC6F
6004D52D D1A086F4 A6703944 A9FC2E0B 9374BD39 5B3FC53E 78A51CE2 59B33FFE
938D9508 74027156 4A6C1367 05B35C3C 7A4BA288 8BF8B5A0 20484626 9FE0EE0B
89C3B819 9BBAEDB1 5322D56C 923B6858 FB068D1D DDF5F1E6 0434DDC3 FC999EAE

E2F906FB AF2C1767 0B9E906C 894CA915 B33C3828 45CEF407 F0615947 2BCABFEE
4DDEADC7 B2E10203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 1459FCC8 38430A05 DED38288 B261BA2D 8DB3E52E
9C301D06 03551D0E 04160414 59FCC838 430A05DE D38288B2 61BA2D8D B3E52E9C
300D0609 2A864886 F70D0101 05050003 82010100 06BC82C7 B0E84AAE 1A12CC08
84252F0E 02088E0D 59C2C6E1 D509708B 2A42D282 258C0C22 9C8B3E9E 9FA60CC1
13E56018 A78A71B1 8E3BDC9A 033E6B65 5435ABF6 04E01500 F5CE30CC F221D554
9C89D156 98B47F42 DDA943B5 489E4769 9B6FDEAC D3CAE60E C87762D2 E7679391
EA5DE4A5 5CF0DFB2 0325A976 586CC08B 8EEA0278 B77D7729 7ECB22C2 ED71F55E
4EC0D811 94E74D18 2DD192C1 1794CD1E F2750596 C0F97D22 9620D9D1 FE929A2D
C7B783C6 0F2D3187 65D67A2B 3B1996AC 3BA2ABB8 B3B7DAEB F93697FB 122A6849
ADDBD880 FF9328D4 67C76337 800C7B4C 2A131A18 C411F63A 032B0281 AB4AE BBB
0942399F 58474747 9F4DB8F1 D2CDBA4C CC39ABC5

quit

!

!

license boot level network-advantage addon dna-advantage

!

!

diagnostic bootup level minimal

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

memory free low-watermark processor 131688

!

username DatacoreAdmin password 7 096B5C060C1546405B5E526E

!

redundancy

mode sso

!

!

!

!

crypto engine compliance shield disable

!

!

transceiver type all

monitoring

!

!

class-map match-any system-cpp-police-ewlc-control

description EWLC Control

class-map match-any system-cpp-police-topology-control

description Topology control

class-map match-any system-cpp-police-sw-forward

description Sw forwarding, L2 LVX data packets, LOGGING, Transit Traffic

class-map match-any system-cpp-default

description EWLC Data, Inter FED Traffic

class-map match-any system-cpp-police-sys-data

description Openflow, Exception, EGR Exception, NFL Sampled Data, RPF Failed

class-map match-any system-cpp-police-punt-webauth

description Punt Webauth

class-map match-any system-cpp-police-l2lvx-control

description L2 LVX control packets

class-map match-any system-cpp-police-forus

description Forus Address resolution and Forus traffic

class-map match-any system-cpp-police-multicast-end-station

description MCAST END STATION

class-map match-any system-cpp-police-high-rate-app

description High Rate Applications

class-map match-any system-cpp-police-multicast

description MCAST Data

class-map match-any system-cpp-police-l2-control

description L2 control

class-map match-any system-cpp-police-dot1x-auth

description DOT1X Auth

class-map match-any system-cpp-police-data

description ICMP redirect, ICMP_GEN and BROADCAST

class-map match-any system-cpp-police-stackwise-virt-control

description Stackwise Virtual OOB

class-map match-any non-client-nrt-class

class-map match-any system-cpp-police-routing-control

description Routing control and Low Latency

class-map match-any system-cpp-police-protocol-snooping

description Protocol snooping

class-map match-any system-cpp-police-dhcp-snooping

description DHCP snooping

class-map match-any system-cpp-police-ios-routing

description L2 control, Topology control, Routing control, Low Latency

class-map match-any system-cpp-police-system-critical

description System Critical and Gold Pkt

class-map match-any system-cpp-police-ios-feature

description

ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTData,Transit,D
OT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusARP,McastEndStn,Openflow,Exc
eption,EGRExcption,NflSampled,RpfFailed

!

policy-map system-cpp-policy

!

!

!

!

!

!

!

!

!

!

!

!

interface Port-channel1

description Link to FL2

switchport trunk native vlan 1000

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000

switchport mode trunk

switchport nonegotiate

ip arp inspection trust

ip dhcp snooping trust

!

interface GigabitEthernet0/0

vrf forwarding Mgmt-vrf

no ip address

negotiation auto

!

interface GigabitEthernet1/0/1

description Link to Host 1

switchport access vlan 10

switchport mode access

switchport port-security maximum 2

switchport port-security violation restrict

switchport port-security mac-address sticky

switchport port-security mac-address sticky 24fb.e389.6c43

switchport port-security mac-address sticky c84d.4427.37d2

switchport port-security

spanning-tree portfast

spanning-tree bpduguard enable

ip dhcp snooping limit rate 10

!

interface GigabitEthernet1/0/2

description Link to Router

switchport trunk native vlan 1000

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000

switchport mode trunk

switchport nonegotiate

ip arp inspection trust

ip dhcp snooping trust

!

interface GigabitEthernet1/0/3

description Link to Host 2

switchport access vlan 20

switchport mode access

switchport port-security maximum 2

switchport port-security violation restrict

switchport port-security mac-address sticky

switchport port-security

spanning-tree portfast

spanning-tree bpduguard enable

ip dhcp snooping limit rate 10

!

interface GigabitEthernet1/0/4

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/5

switchport trunk native vlan 1000

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000

switchport mode trunk

switchport nonegotiate

ip arp inspection trust

channel-group 1 mode active

ip dhcp snooping trust

!

interface GigabitEthernet1/0/6

switchport trunk native vlan 1000

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000

switchport mode trunk

switchport nonegotiate

ip arp inspection trust

channel-group 1 mode active

ip dhcp snooping trust

!

interface GigabitEthernet1/0/7

switchport trunk native vlan 1000

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000

switchport mode trunk

switchport nonegotiate

ip arp inspection trust

channel-group 1 mode active

ip dhcp snooping trust

!

interface GigabitEthernet1/0/8

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/9

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/10

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/11

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/12

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/13

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/14

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/15

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/16

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/17

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/18

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/19

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/20

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/21

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/22

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/23

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/24

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/1/1

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/1/2

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/1/3

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

```
interface GigabitEthernet1/1/4
```

```
switchport access vlan 999
```

```
switchport mode access
```

```
switchport port-security
```

```
ip arp inspection limit rate 1
```

```
shutdown
```

```
spanning-tree bpduguard enable
```

```
ip dhcp snooping limit rate 1
```

```
!
```

```
interface AppGigabitEthernet1/0/1
```

```
switchport access vlan 999
```

```
switchport mode access
```

```
switchport port-security
```

```
ip arp inspection limit rate 1
```

```
shutdown
```

```
spanning-tree bpduguard enable
```

```
ip dhcp snooping limit rate 1
```

```
!
```

```
interface Vlan1
```

```
no ip address
```

```
!
```

```
ip forward-protocol nd
```

```
ip http server
```

```
ip http secure-server
```

```
ip ssh version 2
```

!

!

!

!

!

!

control-plane

service-policy input system-cpp-policy

!

banner motd ^CC Unauthorized Access is Prohibited ^C

!

line con 0

password 7 0908691B1610074328232A

login

stopbits 1

line vty 0 4

password 7 1453351903113A7A121C0A

login local

transport input ssh

line vty 5 31

login

transport input ssh

!

!

!

!

!

!

!

end